

Basic samovar usage

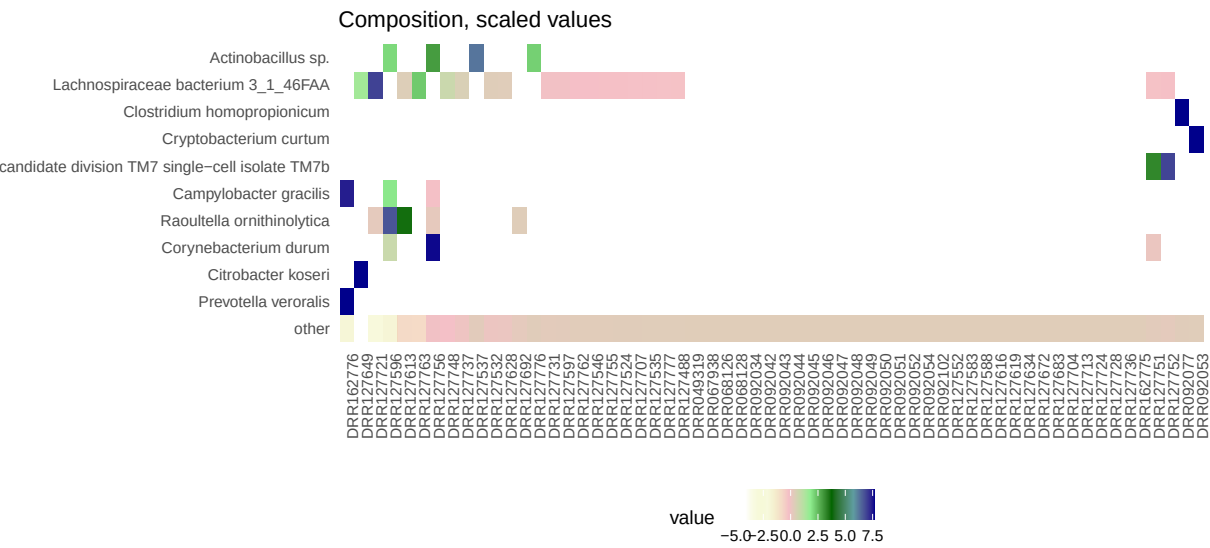
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Download data

Download and vizualize data from GMrepo

```
teatree <- GMrepo_type2data_or_example(number_to_process = 1500)
#> Initial: 873 x 60
#> After filtering: 873 x 60
#> GMrepo unavailable; using packaged example abundance table.
viz_composition(teatree, type = "tile", interactive = F, top = 10)
#> Warning in geom_tile(aes(x = samples, fill = value, y = sp, text = raw_value)):
#> Ignoring unknown aesthetics: text
```



Preprocessing

Filter data

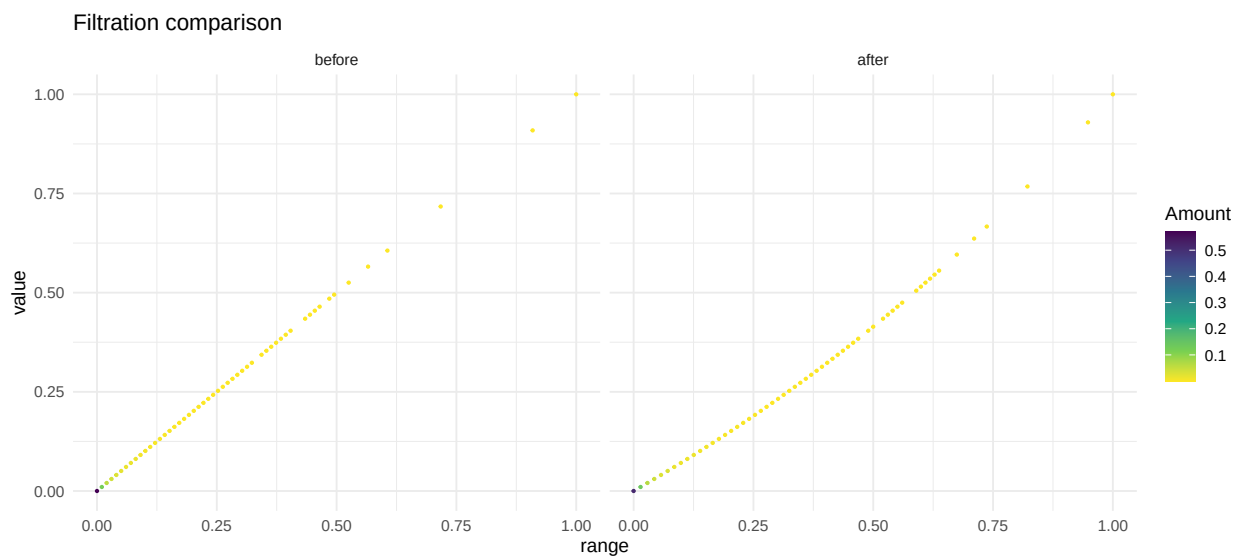
```
tealeaves <- teatree %>%  
  teatree_trim(treshhold_species = 3,  
              treshhold_samples = 3,  
              treshhold_amount = 10-3)  
#> Initial: 871 x 60  
#> After filtering: 177 x 60
```

Normalizing

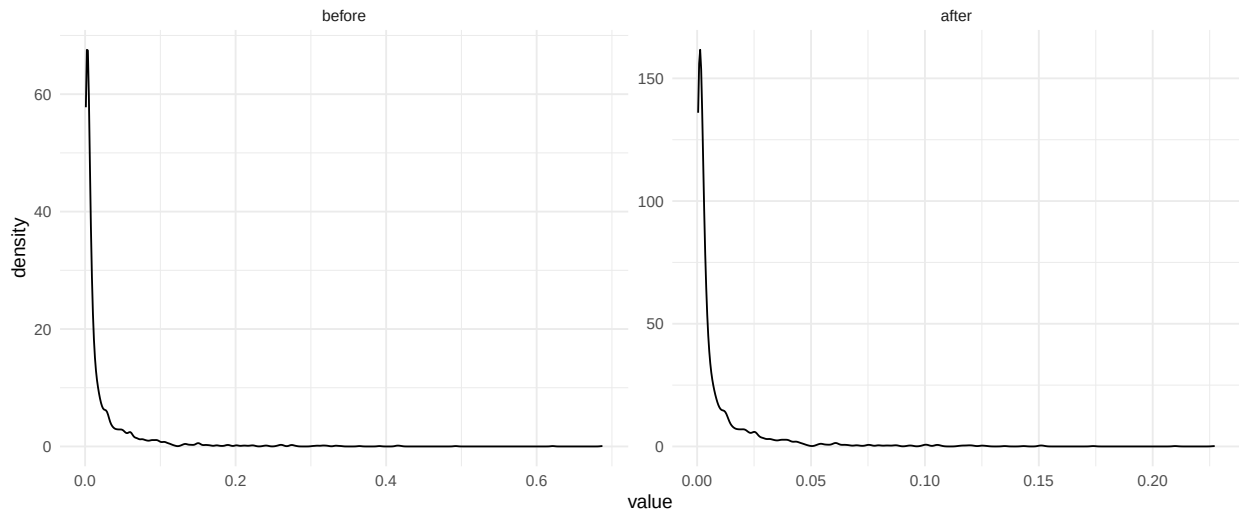
If you build teatree by your own, do rescaling stage when building via `rescale_samovar(teatree)` or assigning `teatree$min_value` and `teatree$max_value` is required

Good approximation to normal distribution is required for `glm` generating methods

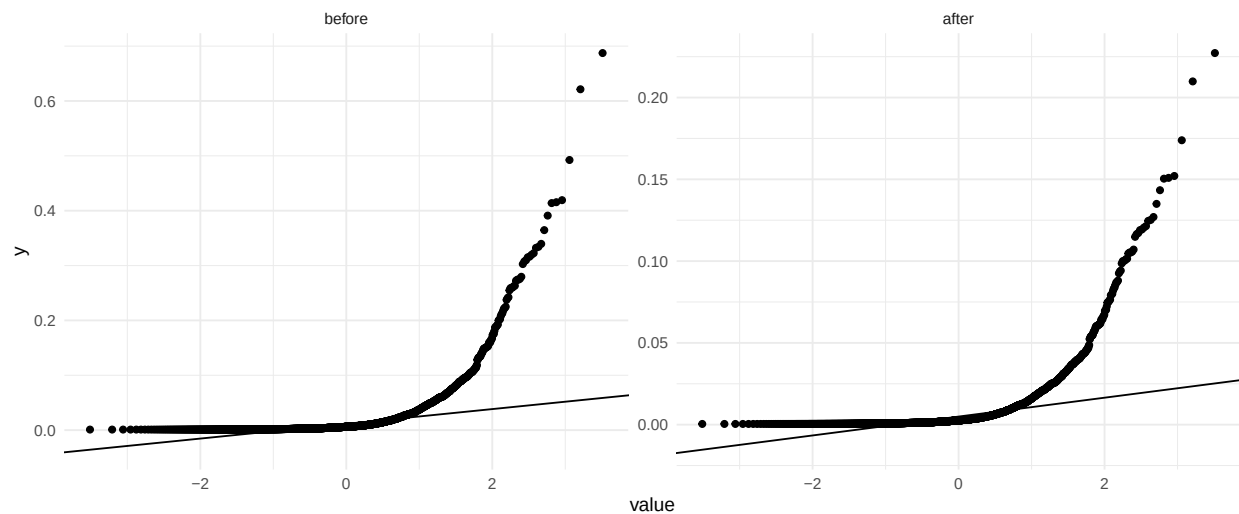
```
teabag <- tealeaves %>%  
  tealeaves_pack()
```



Density plot; only values > 0



QQ-plot; only values > 0

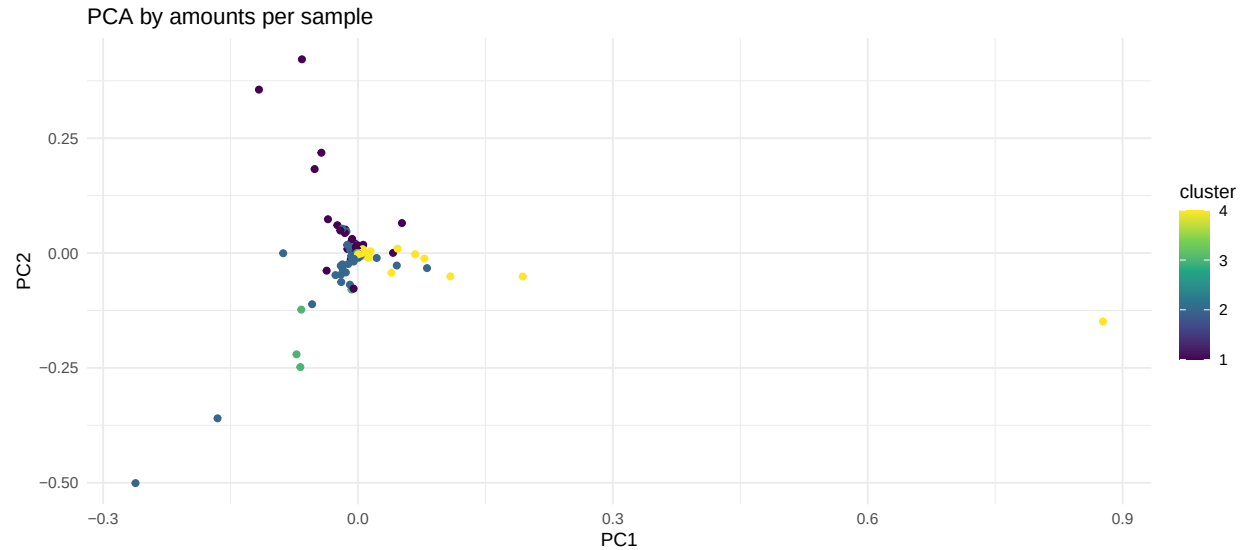


Clustering

Do hierarchical (deprecated) or aggregating clustering of species

Remember: if you want to re-filter, it is better to re-do welding stage to avoid crashes in future!

```
concoction <- teabag %>%
  teabag_brew(min_cluster_size = 40, max_cluster_size = 150)
#> Warning in reducing maximum sample: to similar species!
#> If you want to use bootstrap for samovar build, it might be better to re-filter species
```



Build samovar

```
data.samovar <- concotion %>%
  concotion_pour(probability_calculation = "simple")
#> --- Prepare tea leaves for different clusters ---
#>
#> --- Prepare tea leaves for calculation between clusters ---
#>
#> --- Samovar built ---
```

Generate data

```
new_data <- data.samovar %>%
  samovar_boil(N = 100, avoid_zero_generations = T)
#> --- Generation started ---
#>
#> --- Generation done ---
```

The generated data:

```
viz_composition(new_data, reord_samples = "hcl", type = "tile", interactive = F, top = 10)
#> Warning in geom_tile(aes(x = samples, fill = value, y = sp, text = raw_value)):
#> Ignoring unknown aesthetics: text
```

